

Shared Mental Models in Generative AI-Based Multi-Agent Systems: A Simulation Study Grounded in Human Team Cognition Theory

AUTHOR

LEONARD HEGEMANN

CHAIR OF DIGITAL MANAGEMENT

UNIVERSITY OF HOHENHEIM, GERMANY

I. Introduction

Collective decision making in tasks involving distributed information depends on whether team members disclose and integrate their distributed knowledge. This challenge is central to complex knowledge work, which requires the integration of diverse cognitive skills and is therefore performed collaboratively. Generative artificial intelligence (GenAI) based multi-agent systems (MAS) offer a computational approach to such tasks by giving agents distinct roles and information profiles. However, the mere presence of multiple agents does not guarantee effective and efficient collaboration. Agents must disclose privately held knowledge, retain relevant contributions, and integrate them into a coherent representation of the evolving task. Human team research explains such coordination through the concept of a shared mental model (SMM). By translating the functions associated with task and team mental models into a computational mechanism, this study investigates whether an SMM-inspired structured memory mechanism improves the efficiency and effectiveness of GenAI-based MAS across different levels of access to prior discussion history.

II. Theoretical Background and Hypotheses

Research commonly explains team performance through the input-process-outcome (I-P-O) framework. Inputs comprise the resources and conditions available to a team, processes describe how members transform inputs into outputs, and outcomes capture the results of their collective work. One central input factor is the SMM. It provides team members with a sufficiently compatible understanding of the task, the team, and the current situation. This common cognitive basis helps members anticipate what information others need, connect individual contributions to the collective task, and coordinate their actions. Within the I-P-O framework, SMMs affect two central team processes: communication and cooperation, which in turn shape team performance.

GenAI-based MAS face an analogous collaboration problem when specialised agents must integrate distributed information into a joint decision. To address this problem, this study implements an SMM-inspired structured memory mechanism. It records candidate-specific strengths and concerns, the discussion coverage, the information distribution, agents' current positions, and unresolved issues. It is then reintroduced into the agents' context before subsequent contributions. Its effects may depend on how much prior discussion history remains accessible alongside it. This amount is termed discussion history transparency. The presence of structured memory and the level of discussion history transparency together define the system's context configuration.

Accordingly, the structured memory mechanism is expected to improve communication quality (H1a) and cooperation quality (H1b). Within the structured memory configurations, moderate discussion history transparency is expected to outperform low discussion history transparency because agents can better trace and ground previous contributions (H2a, H2b). Moderate discussion history transparency is also expected to outperform high discussion history transparency because excessive context may reduce the salience of decision-relevant information (H3a, H3b). Finally, higher communication and cooperation quality are expected to improve both efficiency and effectiveness (H4a–H5b).

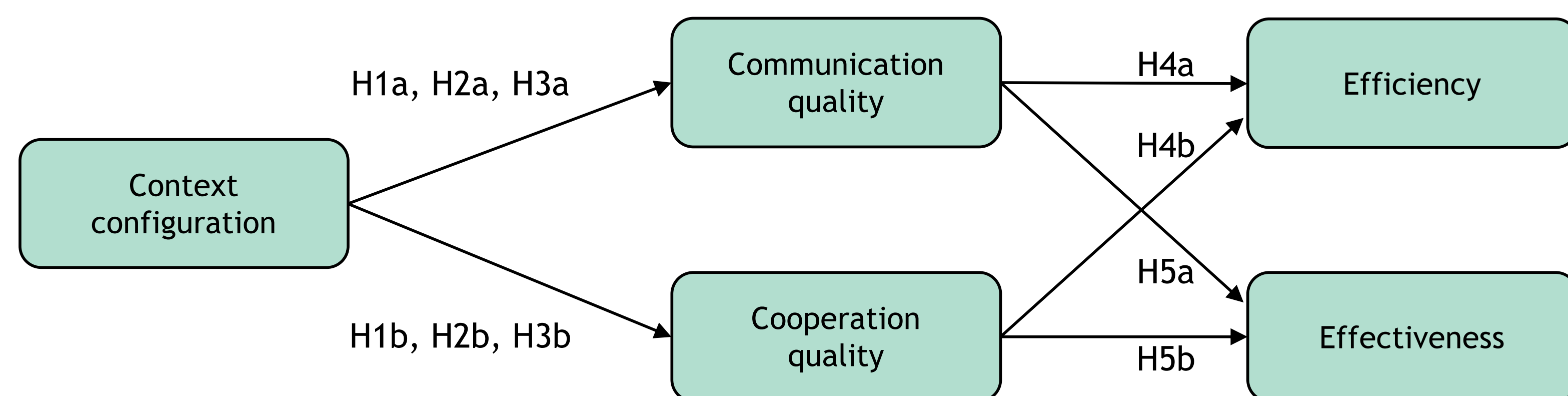


Figure 2: Conceptual research model and hypothesised relationships

III. Methodology

The study is based on 800 independent simulations of a hidden-profile personnel-selection task. Three GenAI-based agents acted as an airline hiring panel and selected one of four candidates for a long-distance pilot position. According to the hidden profile paradigm, all agents received shared task materials, while also holding private information about the candidates. Candidate C was the superior choice in the complete information set and could only be identified if agents disclosed and integrated their private facts. The study compared a baseline without a structured memory mechanism with three structured memory configurations that differed in discussion history transparency: low (only the immediately preceding contribution), moderate (full public discussion history), and high (full public history plus prior reasoning traces). The study measured communication quality, cooperation quality, weighted token usage as an inverse proxy for efficiency, and correct selection of Candidate C as a proxy for effectiveness.

IV. Results

OLS regression models with HC3 standard errors were estimated for communication quality, cooperation quality, and efficiency, while effectiveness was analysed using a logistic regression. The analyses yielded three main findings. First, the structured memory configurations achieved higher percentages of correct decisions than the baseline but required substantially more weighted tokens. Low discussion history transparency produced the highest communication and cooperation scores, and the highest percentage of correct decisions among all context configurations. Second, greater access to discussion history was not necessarily beneficial. Compared with moderate discussion history transparency, high transparency increased weighted token usage and reduced effectiveness. Third, communication and cooperation fulfil distinct functions in GenAI-based MAS. Greater communication quality was associated with a higher probability of a correct decision but also with greater token usage. Greater cooperation quality was associated with lower token usage and showed a marginal positive association with effectiveness. Overall, GenAI-based MAS benefit more from a compact, structured representation of the team's evolving knowledge than from extensive access to prior discussion history. Designers should therefore prioritise structured memory over raw discussion history, as context composition is more important than context volume alone.

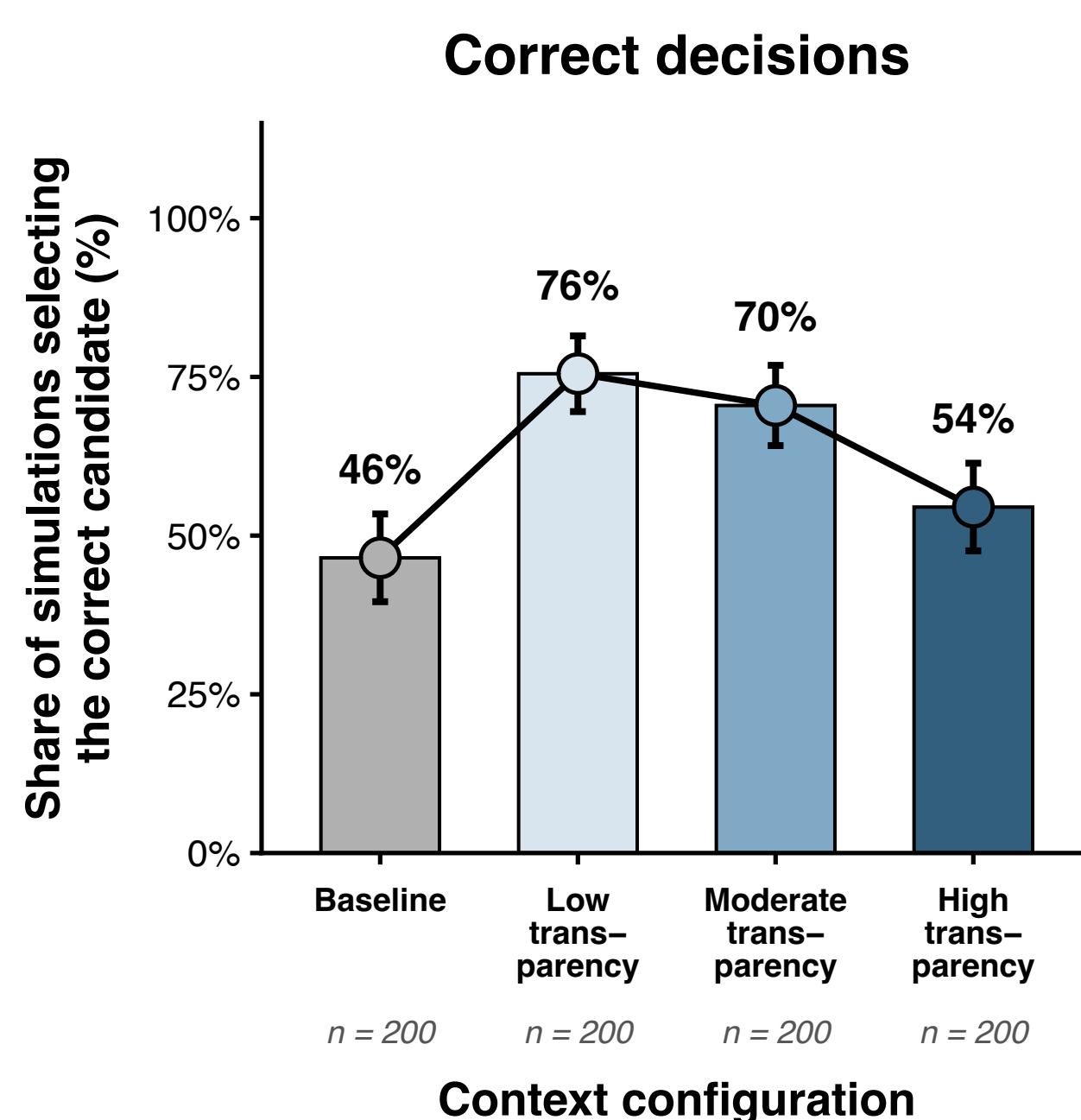


Figure 3: Correct-decision rates across experimental conditions with 95% confidence intervals

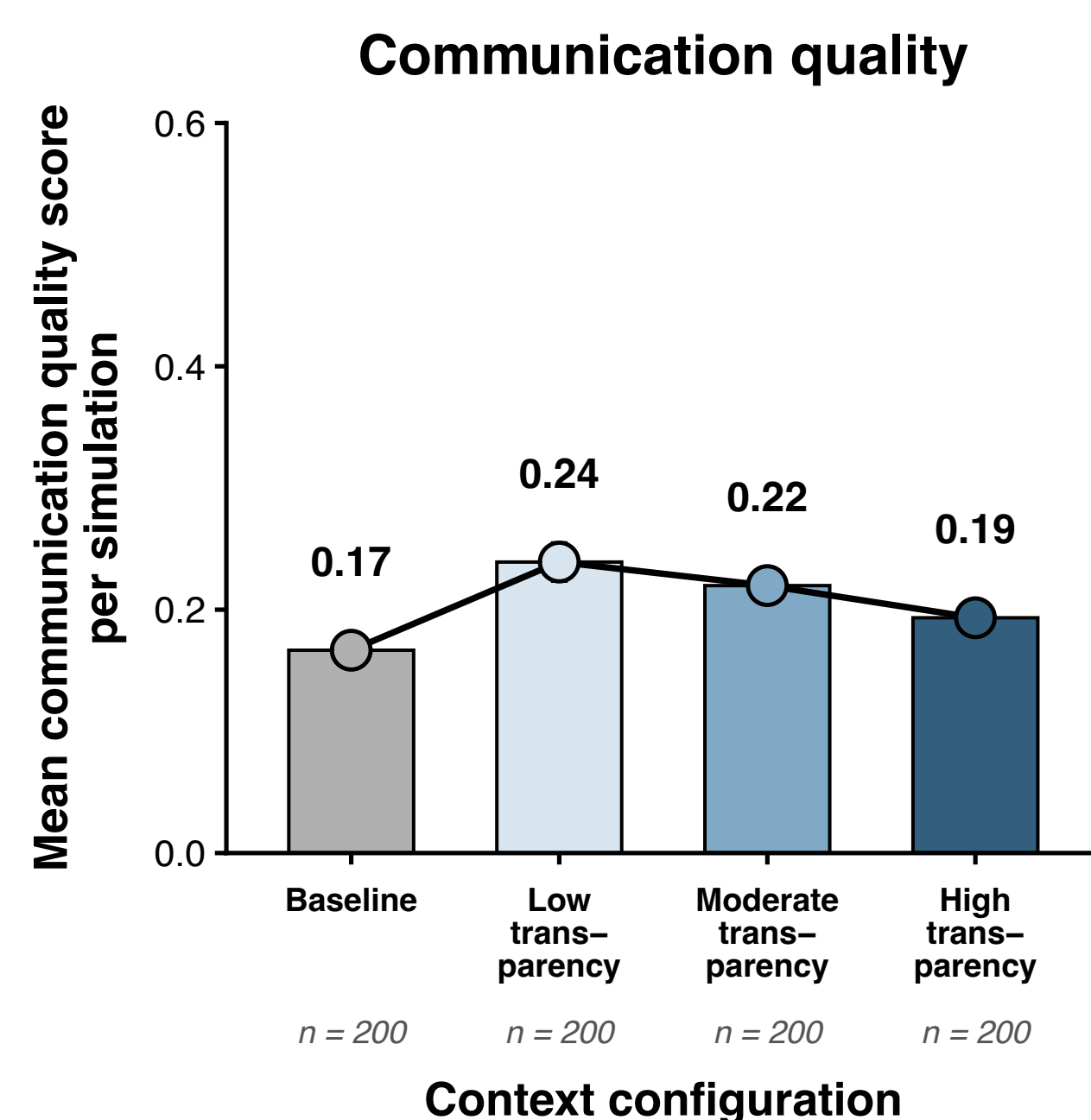


Figure 4: Mean communication quality scores (ranging from 0 to 1) across experimental conditions with 95% confidence intervals

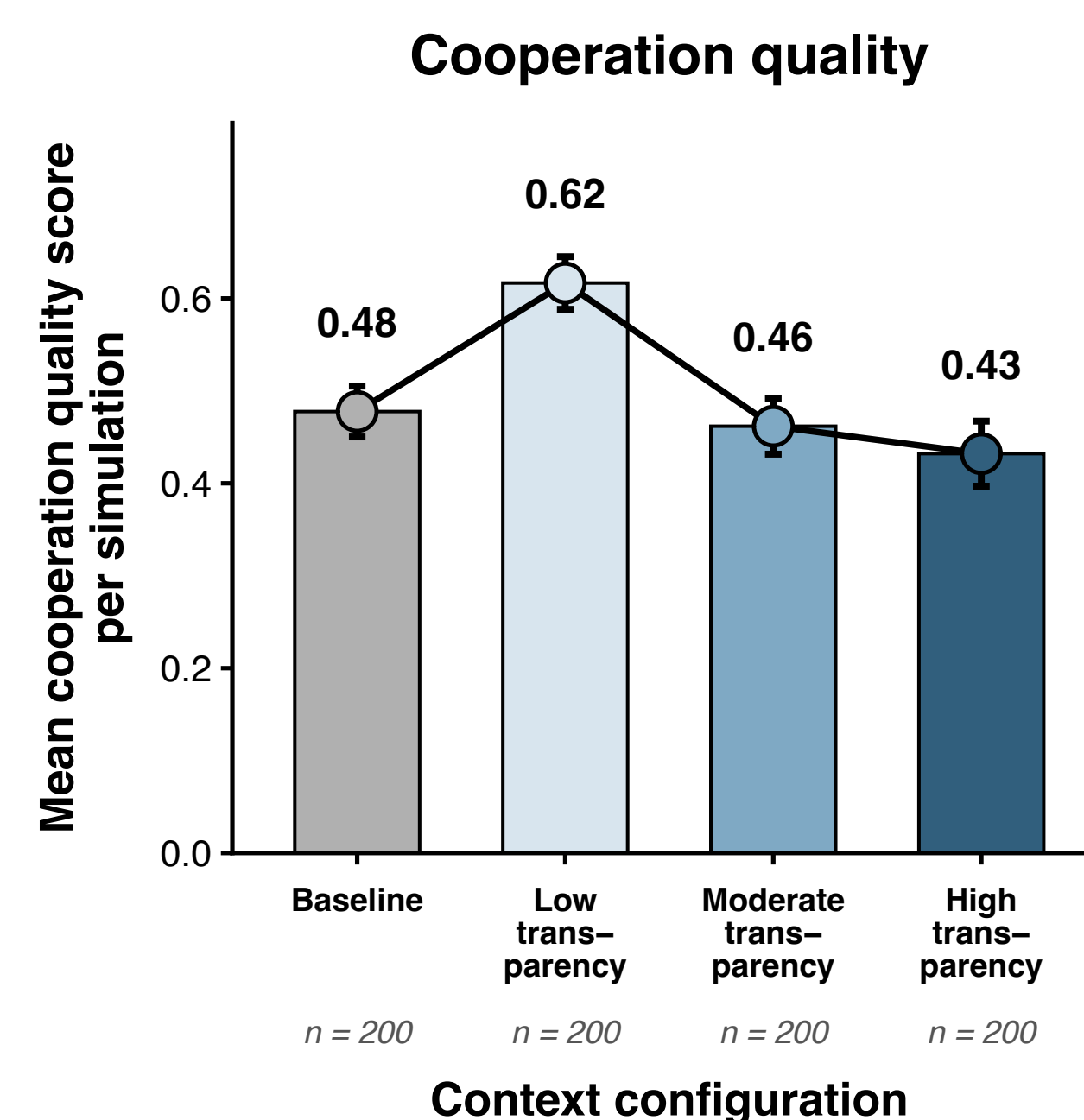


Figure 5: Mean cooperation quality scores (ranging from 0 to 1) across experimental conditions with 95% confidence intervals

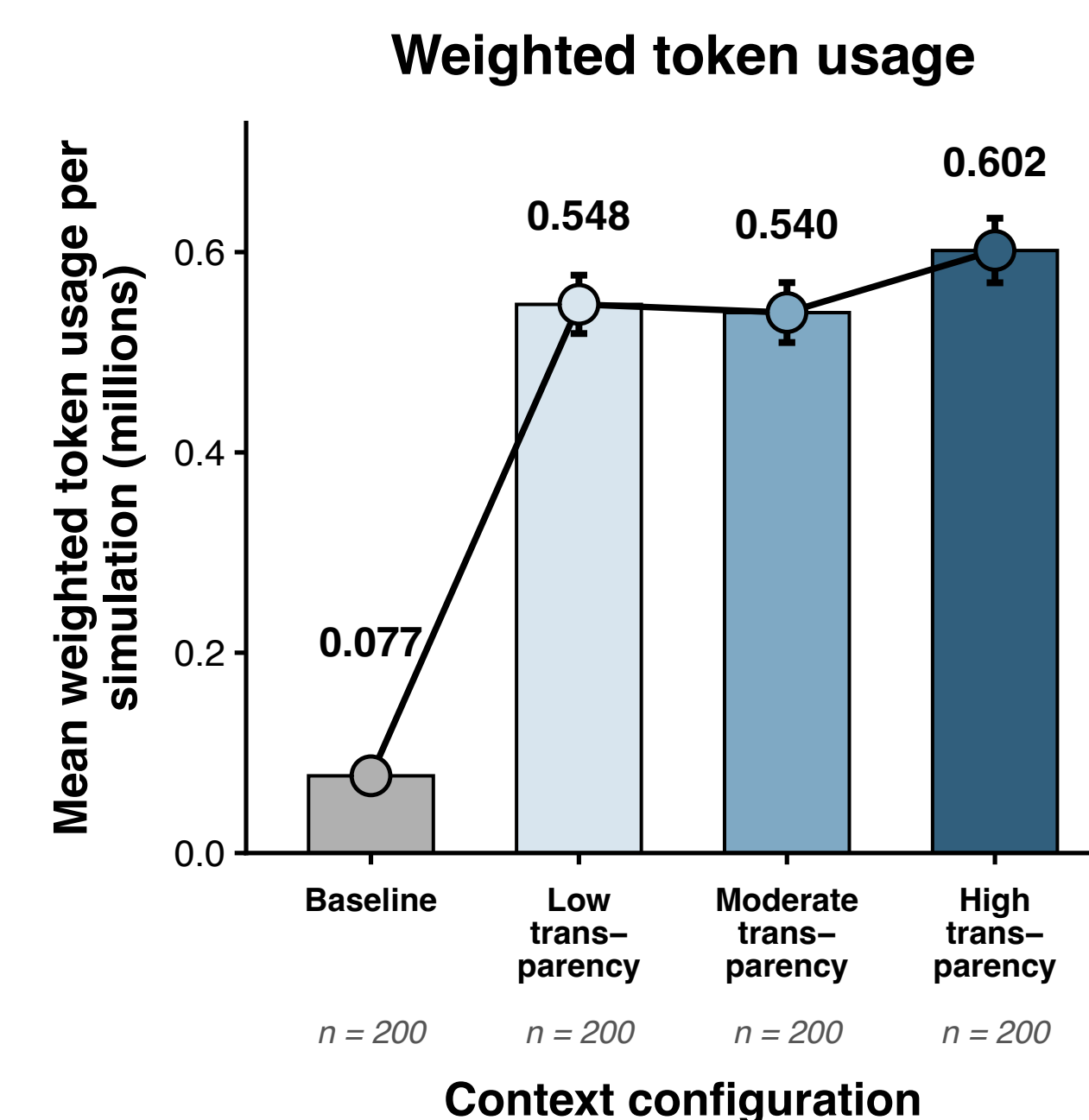


Figure 6: Mean weighted token usage across experimental conditions with 95% confidence intervals



CONTACT

Leonard Hegemann

70794 Filderstadt, Germany

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